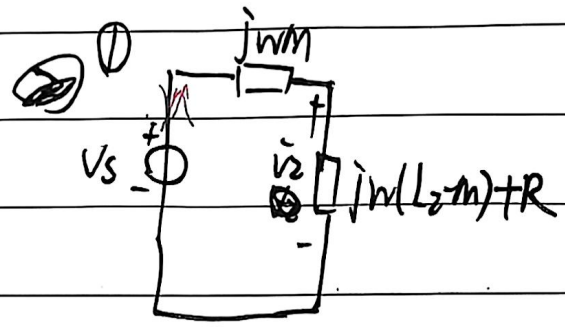
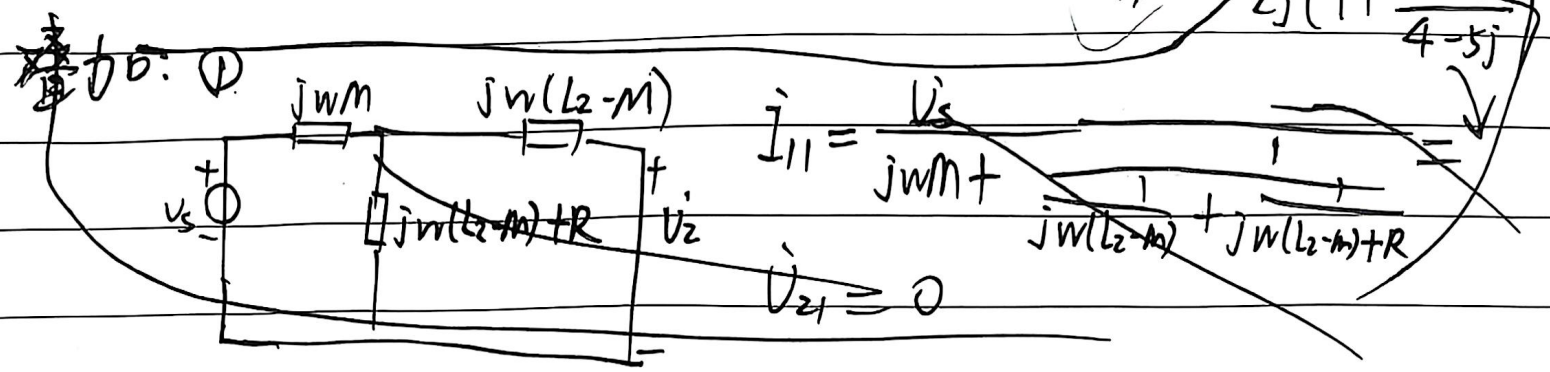
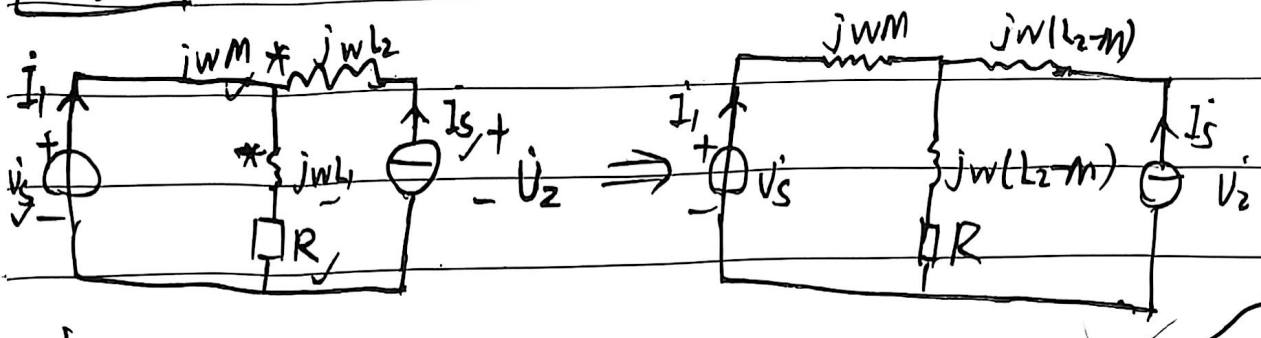


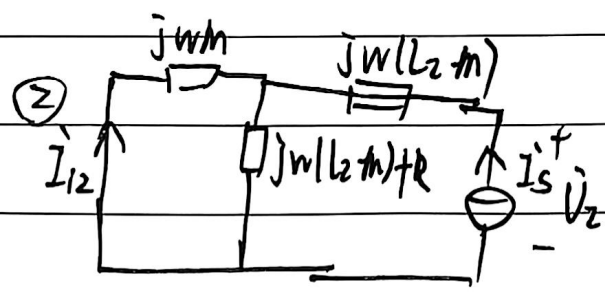
5.34

注意方向 [关联参考方向]



$$\dot{I}_{11} = \frac{\dot{V}_s}{j\omega M + j\omega(L_2 - M) + R} = \frac{20\angle 0^\circ}{4j + 5}$$

$$\dot{U}_{21} = \frac{\dot{V}_s}{j\omega M + j\omega(L_2 - M) + R} [j\omega(L_2 - M) + R] = \frac{20\angle 0^\circ}{4j + 5} (2j + 5)$$



$$\dot{U}_{22} = \dot{I}_s [j\omega(L_2 - M) + \frac{1}{\frac{1}{j\omega(L_2 - M) + R} + \frac{1}{j\omega M}}]$$

$$= 2\angle 90^\circ \cdot \frac{20j - 12}{4j + 5}$$

$$\dot{I}_{12} = 0$$

$$\dot{I}_s = \frac{2j + 5}{4j + 5}$$

$$\therefore \textcircled{1} \dot{I}_1 = \frac{16 + 10j}{4j + 5} = \frac{24 - 10j}{4j + 5}$$

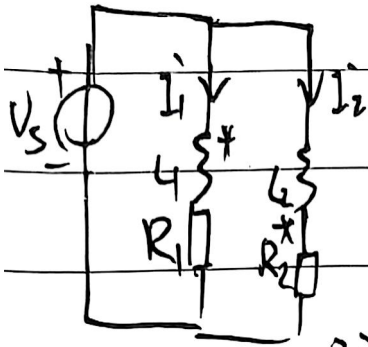
$$\dot{U}_2 = \frac{16j + 60}{4j + 5}$$

[计算有错]

$$P_0 = \text{Re}[\dot{V}_s^* \dot{I}_1] = \frac{2400}{\sqrt{41}} \quad 39W \text{ 发出}$$

$$P_1 = \text{Re}[\dot{I}_s^* \dot{U}_2] = -\frac{3200}{\sqrt{41}} \quad 7.8W \text{ 吸收}$$

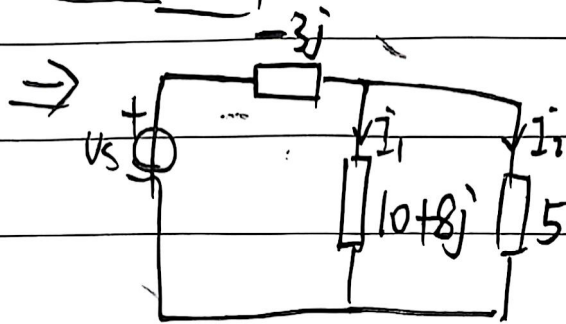
5.35



$$P_{w1} = \text{Re}[\dot{I}_1 \dot{V}_s^*] \quad P_{w2} = [\dot{I}_2 \dot{U}_s^*]$$

$$\dot{V}_s = 100 \angle 0^\circ, \quad \omega = 1000$$

$$j\omega L_1 = 5j \quad j\omega L_2 = 13j \quad j\omega L_3 = 3j$$



$$\dot{U}' = (-3j) \cdot \frac{\dot{V}_s}{-3j + \frac{1}{\frac{1}{10+8j} + \frac{1}{5+16j}}} = 30.78 - 45.26j$$

$$\dot{I}_1 = \frac{\dot{U}'}{10+8j} = \frac{30.78 - 45.26j}{164} (10 - 8j) =$$

$$\dot{I}_2 = \frac{\dot{U}'}{5+16j} = \frac{30.78 - 45.26j}{281} (5 - 16j) =$$

$$P_{w1} = \text{Re}[\dot{U}'^* \dot{I}_1] = \frac{(30.78)^2 + (45.26)^2}{164} 10 = 522.8 \text{ W} \quad \text{吸收}$$

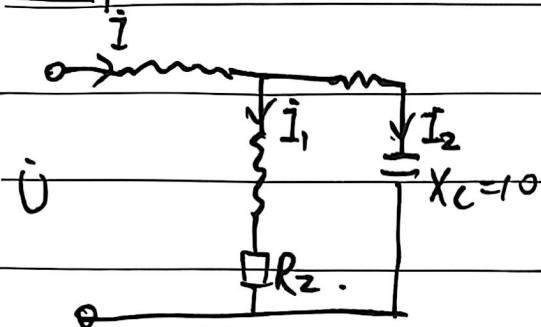
$$P_{w2} = \text{Re}[\dot{U}'^* \dot{I}_2] = 152.6 \text{ W} \quad \text{吸收}$$

等效电路对内不成立

功率表读数仍为 $U=100$

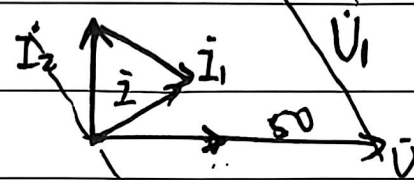
☆：等效对内不成立

5.40

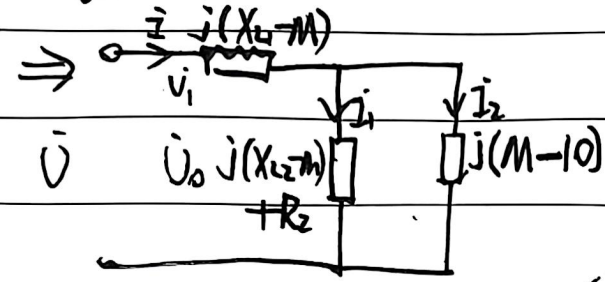
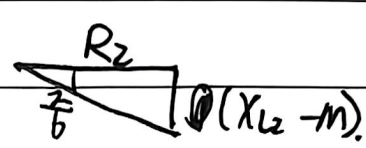


$$I_1^2 R_2 = 433 \text{ W} \Rightarrow R_2 = 4.33 \Omega$$

不妨设 $m = 10 \Omega$



$$\varphi_1 = \frac{\pi}{6}$$



$$\therefore X_{L2} - m = \frac{1}{\sqrt{3}} R_2 = 2.5 \Omega, \quad I_1 = 10 \angle -\frac{\pi}{6}$$

$$\therefore \vec{U} = \vec{I}_1 (4.33 + j2.5) = 10 \left(\frac{\sqrt{3}}{2} - \frac{j}{2} \right) (4.33 + j2.5)$$

$$\vec{U} = \vec{I}_1 (4.33 + j2.5) = 50 \angle 0^\circ$$

$$\therefore j(m - 10) = \frac{\vec{U}}{\vec{I}_2} = 5 \angle -\frac{\pi}{2} = -5j \quad \therefore m = 5 \Omega$$

$$\therefore \vec{I} = 10 \angle \frac{\pi}{6}, \quad \vec{U}_1 = 50 \angle \frac{\pi}{3}$$

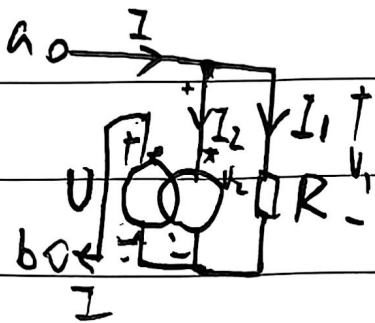
$$\therefore j(X_{L1} - m) = \frac{\vec{U}_1}{\vec{I}} = 5 \angle \frac{\pi}{2} = 5j$$

$$\therefore X_{L1} = m + 5 = 10 \Omega$$

or $U_2 = 0, \quad X_{L1} = 5 \Omega$

15.42

$$Z_{\Delta} = \frac{V_{AB}}{i}$$



$$\underline{I = I_1 + I_2}$$

$$\textcircled{D} N_1 I = N_2 I_2$$

$$V_1 = I_1 R$$

$$U_2 = U_1$$

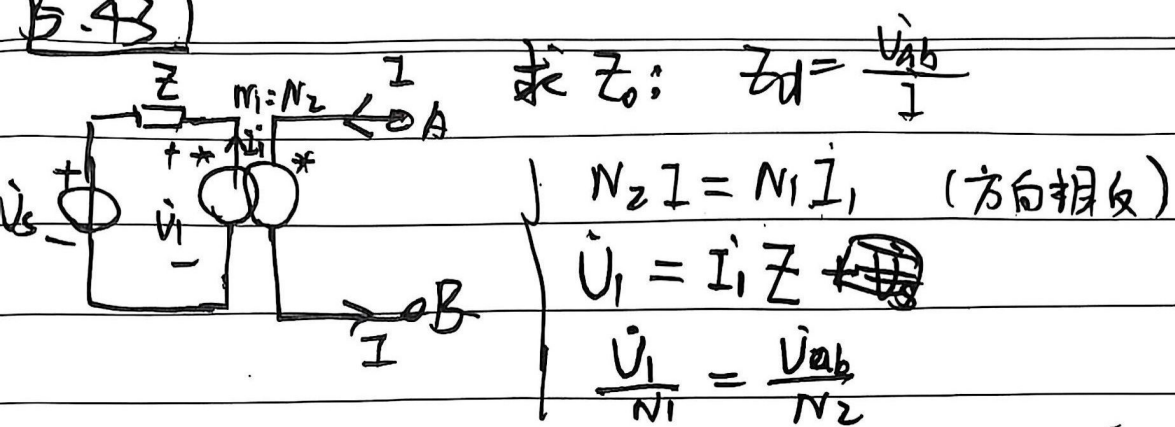
$$\frac{V_1}{N_1} = \frac{V_2}{N_2}$$

$$U_{ab} = U_1 - U$$

$$\Rightarrow V_{ab} = \left(1 - \frac{N_1}{N_2}\right) \left(1 - \frac{N_1}{N_2}\right) I R$$

$$\therefore Z_{\text{eff}} = \left(1 - \frac{N_1}{N_2}\right)^2 R$$

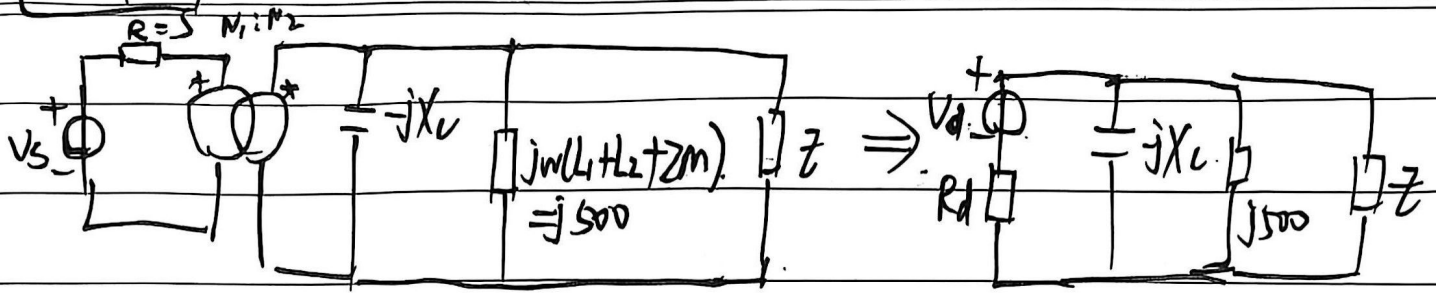
5.43



$$\Rightarrow \dot{U}_{ab} = \frac{N_2}{N_1} \left[\frac{N_2}{N_1} \dot{I} Z \right] \Rightarrow Z_d = \left(\frac{N_2}{N_1} \right)^2 Z$$

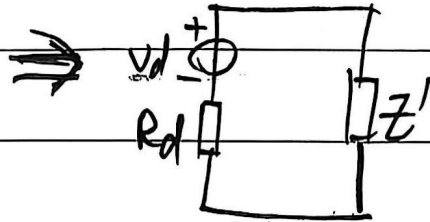
$$\dot{U}_{ab} = \frac{N_2^2}{N_1^2} \dot{U}_s$$

5.44



$$\dot{V}_d = \left(-\frac{N_2}{N_1}\right) \dot{V}_s$$

$$R_d = \left(\frac{N_2}{N_1}\right)^2 R$$



$$P_{Z'} = P_Z$$

$$\therefore \frac{1}{Z'} = \frac{1}{R_d} = j\frac{1}{X_c} - j\frac{1}{500} + \frac{1}{80 + j60} = 0.008 + j\left(\frac{1}{X_c} - \frac{1}{500} - \frac{3}{500}\right)$$

$$\Rightarrow X_c = 125 \Omega$$

$$\frac{N_1}{N_2} = 0.2$$

$$\dot{V}_d = 5 \dot{V}_s = 50 \angle 0^\circ, R_d = 125 \Omega$$

$$\Rightarrow \therefore P_Z = \frac{V_d^2}{4R_d} = 5 \text{ W}$$